Google 🔍 Ayush Singh - distributed systems engineer ✕ ⚙ ⋮⋮⋮ You

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✦ AI Overview · Executive Summary

**Software Engineer III (L4)** at **Google** (Munich) with 4+ years building planet-scale **distributed systems**, **low-latency** C++/Java RPC frameworks, fault-tolerant distributed file systems, and real-time streaming pipelines across Google, Komprise, Accenture, and Tookitaki.

- **Planet-Scale Core Infra:** Defaulted architecture across ~5,200 microservice nodes (Search, YouTube, Maps) & built distributed storage serving 100,000+ services.
- **High-Impact Systems ROI**: Delivered $2.5M+/quarter savings (SLA platform) and 94% streaming latency reduction (850ms → <50ms).
- **Fast Track Record**: Promoted L3 → L4 in 15 months; VP Spot Bonus ; 3× Readability (C++, TS, API) & 1040+ PRs reviewed.

## Top stats

| ~5,200 | 100K+ | $2.5M+ | 94% |
|---|---|---|---|
| Prod microservice nodes running my default design | Google services backed by my distributed FileSystem | Quarterly savings via SLA/SLO platform | Streaming latency cut (850ms → <50ms) |

## Work Experience · 4+ Years in Distributed Systems, Storage & Backend

### Google — Software Engineer III (L4), Distributed Systems & Core Infrastructure

**Munich, Germany · Nov 2024 – Present** · C++ · Java · Kotlin · gRPC · Distributed Microservices · Edge Routing

- **Core C++ & Java Request Frameworks:** Shipped global request-path processors into **Google's core Java backend framework (powering all Google products)** and **core C++ Search serving engine**, sustaining 7 billion+ daily requests across Search, YouTube, and Maps.
- **SignedOut Architecture Across 5,200+ Nodes**: Standardized the Google wide SignedOut request-processing architecture defaulted across **~5,200 production microservice nodes** (210+ modules), cutting signal-processing complexity by **40%** and retiring **1,300+ legacy LOC**.
- **Cross-Product Telemetry & Batch Compute:** Built parallel stream shadow-diffing across **YouTube, Play, Search & Payments** (100% live cut-over parity) and migrated **7 critical identity pipelines** to distributed batch compute, reducing p99 serving tail latency by 12 and reclaiming GCUs.
- **High-Velocity Regulatory Rollouts:** Led end-to-end **UK Online Safety Act & US AppStore** distributed backend launches across Web, Android & iOS in 4 weeks across 5+ teams ( VP Spot Bonus ).

### Google — Software Engineer II → III (L4), Distributed Storage & Observability

**Hyderabad, India · Jul 2022 – Nov 2024** · C++ · Go · FUSE · Distributed File Systems · Paxos Consensus · OpenTelemetry

- **Distributed File System (100K+ Services):** Engineered fault-tolerant storage and replication for **Google's internal HDFS-like distributed file system**, optimizing mission-critical binary delivery across **100,000+ internal services**.
- **Overload Protection & Consensus Locks:** Shipped server-side admission control & load-shedding, semaphore RPC concurrency limiting in the **FUSE client**, and migrated distributed snapshot locks off Paxos lock services, cutting lock contention by 65.
- **$2.5M+ Quarterly Savings via SLA/SLO Platform:** Architected a highly available observability & automated **SLA/SLO compliance platform** correlating distributed traces and metrics, saving **$2.5M+ per quarter**.
- **OpenTelemetry Collector & Go Gateway:** Co-architected a trace collector over **gRPC-Web** & encrypted Go asset gateway (50,000+ req/sec); led Chrome 3P cookie deprecation analysis across 120+ services (approved by Google CEO) & promoted to L4 in



# Ayush Singh

Distributed Systems Engineer · Google L4

Overview | Experience | Skills

**Current:** SWE III (L4) at Google

**Experience:** 4+ Years

**Location:** Munich, Germany

**Email:** apparently.ayush@gmail.com

**Phone:** +49 1735 34 8915

**Education:** NIT Agartala · B.Tech EE (CGPA: 9.32 / 10)

**Code Authority:** 3× Google Readability (C++, TS, API) · 1,040+ Reviews

**Honors:** VP Spot Bonus · 7× Peer Bonuses · GFG Rank #72

## Profiles & Links

LinkedIn | Portfolio | GitHub | Email

## Technical skills

Languages

C++ Java Go TypeScript Python Scala Kotlin SQL

Distributed Systems & Storage

gRPC Protobuf Distributed FS (HDFS / FUSE) Spanner Paxos / Raft Consensus Apache Spark Kafka ActiveMQ Snowflake Redis

Cloud, Observability & AI Infra

Kubernetes Docker OpenTelemetry GCP AWS Terraform CI/CD WebGPU / LLM Agents

15 months.

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### Komprise — Software Engineer Intern (Distributed Storage Infrastructure)

**Bengaluru, India · Jan 2022 – Jun 2022** · Python · Multi-Threading · Distributed Systems Testing · CI/CD

- **Parallel Distributed Test Execution:** Built a multi-threaded automation framework executing in parallel across distributed storage nodes, cutting build-verification runtime by **87.5%** and increasing bug detection coverage by **60%**.

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### Accenture — Data Engineer Intern (Cloud Data Infrastructure)

**India · May 2021 – Jul 2021** · AWS S3 · Snowflake · Distributed ETL · Python · SQL

- **10TB+ High-Volume Cloud ETL:** Designed optimized distributed ETL pipelines transferring **10TB+ datasets from AWS S3 to Snowflake** with real-time analytics automation cutting reporting time by **70%**

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### Tookitaki — Software Engineer Intern (Real-Time Distributed Streaming)

**Singapore · Oct 2020 – Jan 2021** · Apache Spark Structured-Streaming · Scala · ActiveMQ · JMS

- **94% Streaming Latency Reduction (850ms → <50ms):** Built a real-time **Apache Spark Structured-Streaming + ActiveMQ** pipeline with custom connection-pooled JMS clients and **exactly-once fault-tolerant recovery**.

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## ✦ Selected Open-Source & Agentic Systems

### 1. YTM AI Host — Real-Time (<100ms) AI Radio Jockey & Audio Streaming Engine · GitHub

TypeScript · Python · Gemini API · WebLLM / WebGPU · On-Device Inference · Real-Time Audio Pre-Fetching

- **Sub-100ms Streaming AI Architecture:** Built a modular local streaming server and Chrome extension for **YouTube Music & Spotify**, combining **on-device WebGPU/WebLLM inference**, multi-engine TTS (Kokoro JS, XTTS v2), and intelligent pre-fetching/caching for **<100ms real-time audio transitions**.

### 2. LinkinJMS (High-Performance Spark Queue Connector) & Multi-Agent Incident Triage

- **LinkinJMS (Scala / Apache Spark / ActiveMQ):** Open-source bidirectional streaming connector achieving **<50ms message latency** with **exactly-once delivery**, AMQP/TLS encryption, and automatic failover.
- **Autonomous Multi-Agent Oncall Triage (Google):** Production Python multi-agent system that fans out parallel diagnostic queries across code changes, rollout logs, and time-series metrics to email root-cause reports before oncallers open the alert.

## People also ask

**What scale of distributed systems does Ayush build at Google?**
His C++/Java middleware runs on the critical request path of Search, YouTube & Maps (~5,200 microservice nodes), and his distributed storage systems serve 100K+ services.

**How fast was Ayush promoted at Google?**
Promoted L3 → L4 (SWE III) in 15 months with a VP of Engineering Spot Bonus, 7× peer bonuses, and 3× Google Readability certifications (C++, TypeScript, API; 1,040+ code reviews).

**What measurable systems impact has Ayush delivered?**
$2.5M+/quarter saved via automated SLA/SLO enforcement, 94% streaming latency reduction (850ms → <50ms), 40% lower RPC complexity, and 87.5% faster distributed test execution.

**Where did Ayush study and what are his algorithmic credentials?**
NIT Agartala (B.Tech EE, 9.32/10 CGPA). Ranked #1 in Institute & #72 Worldwide on GeeksForGeeks (top 0.07% of 100K+ developers), IASc Fellow, and Founding Google DSC Lead.

## People also search for

Distributed File Systems (HDFS/FUSE) High-Throughput gRPC Apache Spark Streaming OpenTelemetry Paxos & Consensus

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